

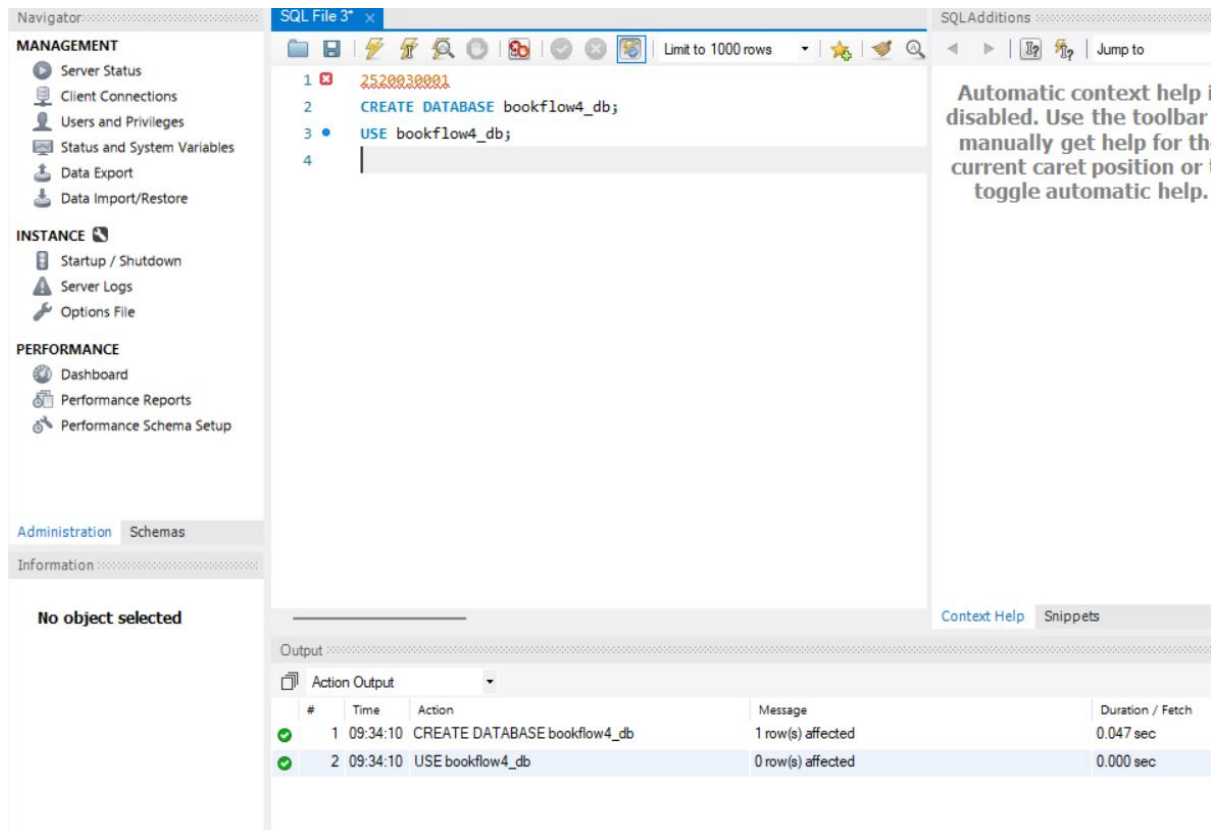
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030001

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Query 1: -CREATE DATABASE



The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'Navigation' tree with categories like MANAGEMENT, INSTANCE, and PERFORMANCE. The central pane shows the SQL File 3* editor with the following queries:

```
1 2520030001
2 CREATE DATABASE bookflow4_db;
3 USE bookflow4_db;
4
```

The right pane shows the 'SQLAdditions' tab with a message: 'Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.'

The bottom pane shows the 'Output' window with the 'Action Output' tab selected. It displays the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
1	09:34:10	CREATE DATABASE bookflow4_db	1 row(s) affected	0.047 sec
2	09:34:10	USE bookflow4_db	0 row(s) affected	0.000 sec

Query 2: Create Books Table (DDL + Constraints)



The screenshot displays the SQL File 3* editor with the following query:

```
1 2520030001
2 CREATE TABLE BOK (
3     book_id INT PRIMARY KEY,
4     title VARCHAR(100) NOT NULL,
5     isbn VARCHAR(20) UNIQUE,
6     published_year INT CHECK (published_year < 2027)
7 );
```

The right pane shows the 'SQLAdditions' tab with a message: 'manually get help for the current caret position or toggle automatic help.'

Output: -

✓	4	09:38:44	CREATE TABLE Bok (book_id INT PRIMARY KEY...	0 row(s) affected	0.109 sec
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Query 3: Insert Books and display

```
11 ✖ 2520030001
12 INSERT INTO Bok (book_id,title,isbn,published_year) VALUES
13 (1,'The Great Bay','9780743273565',2000),
14 (2,'The Indian Culture',2345987456,2010),
15 (3,'1947','9456783457934',2020);
16 • SELECT * FROM Bok;
```

Automatic context disabled. Use the manually get help current caret position toggle automatic

Result Grid

book_id	title	isbn	published_year
1	The Great Bay	9780743273565	2000
2	The Indian Culture	2345987456	2010
3	1947	9456783457934	2020
NULL	NULL	NULL	NULL

Form Editor

Field Types

Output:-

✓	12	12:10:49	INSERT INTO Bok (book_id,title,isbn,published_y...	3 row(s) affected	Records: 3 Duplicates: 0 Wami...	0.062 sec
✓	13	12:10:49	SELECT * FROM Bok LIMIT 0, 1000	3 row(s) returned		0.000 sec / 0.000 sec

Query 4: Members Table: Create

```
CREATE TABLE Members(
member_id INT PRIMARY KEY,
full_name VARCHAR(100),
email VARCHAR(100) UNIQUE
);
```

Output:-

✓	19	12:20:46	CREATE TABLE Members(member_id INT PRIM...	0 row(s) affected		0.063 sec
---	----	----------	--	-------------------	--	-----------

Query 5:- Insert Members table

```
INSERT INTO Members (member_id,full_name,email) VALUES
(101, 'Aswitha','aswitha@gmail.com'),
(102,'Adhritha','adhritha@gmail.com'),
(103,'Manasvi','manasvi@gmail.com');
```

Output: -

20	12:22:31	INSERT INTO Members (member_id,full_name,e...	3 row(s) affected Records: 3 Duplicates: 0 Wami...	0.016 sec
----	----------	---	--	-----------

Query 6: - : Display Members table

```
SELECT * FROM Members;
```

Output: -

The screenshot displays a database management interface. At the top, there's a 'Result Grid' tab. Below it, a table with 3 rows and 3 columns is shown. The columns are 'member_id', 'full_name', and 'email'. The rows contain the following data: (101, Aswitha, aswitha@gmail.com), (102, Adhritha, adhritha@gmail.com), and (103, Manasvi, manasvi@gmail.com). To the right of the table, there are icons for 'Result Grid', 'Form Editor', and 'Field Types'. Below the table, there's a 'Members 2' tab with 'Apply' and 'Revert' buttons. At the bottom, the 'Output' section shows the execution of the query 'SELECT * FROM Members LIMIT 0, 1000', which returned 3 rows in 0.000 seconds.

member_id	full_name	email
101	Aswitha	aswitha@gmail.com
102	Adhritha	adhritha@gmail.com
103	Manasvi	manasvi@gmail.com

Output

20	12:22:31	INSERT INTO Members (member_id,full_name,e...	3 row(s) affected Records: 3 Duplicates: 0 Wami...	0.016 sec
21	12:24:30	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query 7: Create Loans Table

2520030001-Aswitha

```
) CREATE TABLE Loan (  
  loan_id INT PRIMARY KEY,  
  member_id INT,  
  book_id INT,  
  loan_date DATE,  
  FOREIGN KEY (member_id) REFERENCES Members(member_id),  
  FOREIGN KEY (book_id) REFERENCES Bok(book_id)  
);
```

Output:-

```
23 12:30:52 CREATE TABLE Loan (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN... 0 row(s) affected
```

Query 8: Insert 10 Loans

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

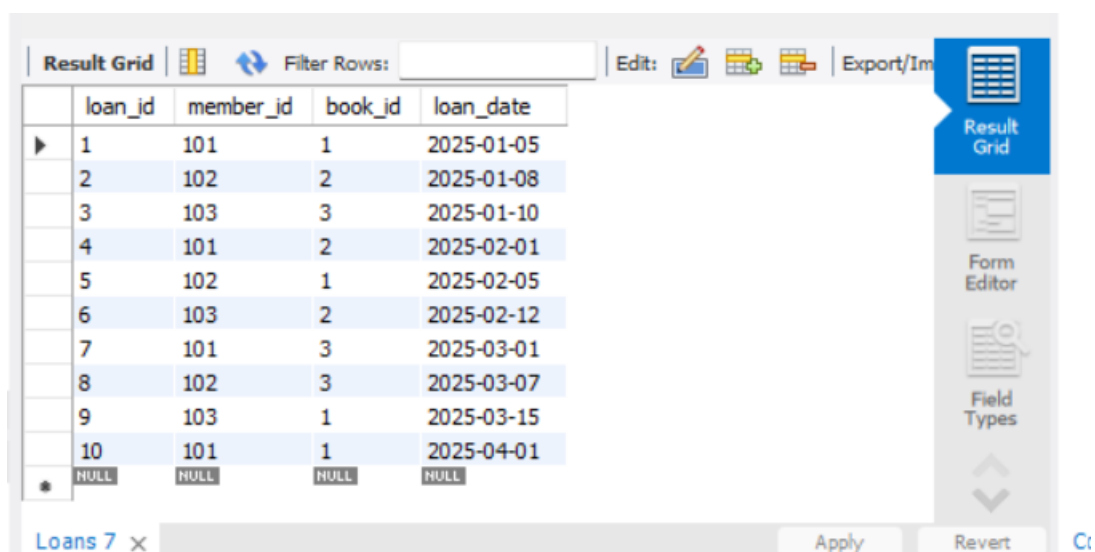
Output: -

```
91 13:40:13 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), ... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 0.04
```

Query 9: Display

```
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');  
SELECT * FROM Loans;
```

Output :-

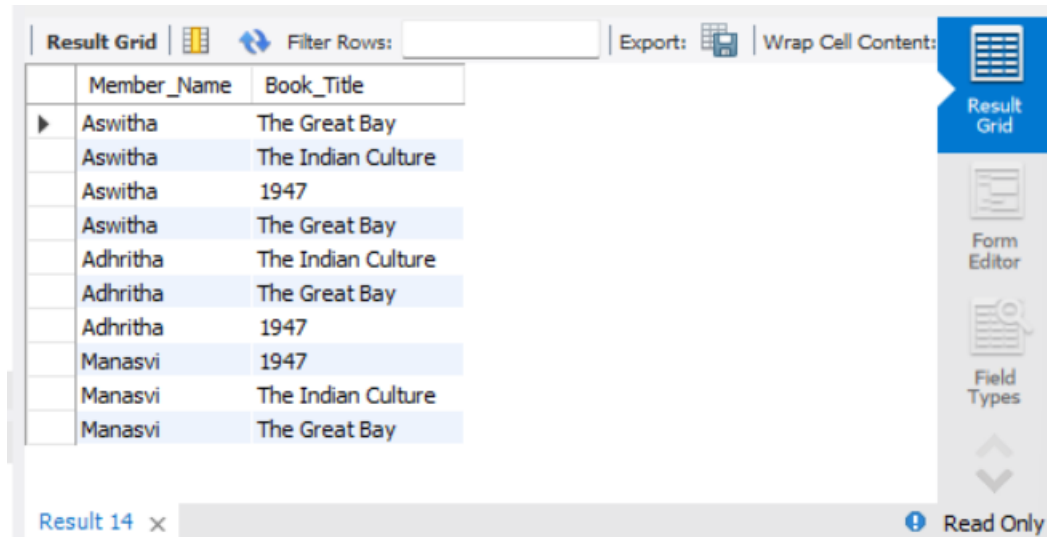


	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
✱	NULL	NULL	NULL	NULL

Query 10: JOIN Query

```
45 • SELECT m.full_name AS Member_Name,b.title AS Book_Title
46 FROM loans l
47 INNER JOIN Members m ON l.member_id = m.member_id
48 INNER JOIN Bok b ON l.book_id = b.book_id;
```

Output: -



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays 10 rows of data with two columns: 'Member_Name' and 'Book_Title'. The data is as follows:

Member_Name	Book_Title
Aswitha	The Great Bay
Aswitha	The Indian Culture
Aswitha	1947
Aswitha	The Great Bay
Adhritha	The Indian Culture
Adhritha	The Great Bay
Adhritha	1947
Manasvi	1947
Manasvi	The Indian Culture
Manasvi	The Great Bay

At the bottom of the grid, it says 'Result 14' and 'Read Only'.

```
✓ 103 13:50:44 SELECT m.full_name AS Member_Name,b.title A... 10 row(s) returned
```

Query 11: GROUP BY Query

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Bok
GROUP BY published_year
ORDER BY published_year;
```

Output :-

Result Grid			Filter Rows:	Export:	Wrap Cell Contents:
	published_year	Total_Books			
▶	2000	1			
	2010	1			
	2020	1			

✓	103	13:50:44	SELECT m.full_name AS member_name, b.title AS...	10 row(s) returned	0.000 sec / 0.000 sec
✓	104	13:54:20	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned	0.016 sec / 0.000 sec

Query 12: Donation via TRANSACTION

```
2520030001-Aswitha
CREATE TABLE Donation_History (
donation_id INT PRIMARY KEY,
book_id INT,
donar_name VARCHAR(100),
donation_date DATE,
FOREIGN KEY (book_id) REFERENCES Bok(book_id)
);
```

Output: -

✓	107	13:57:39	CREATE TABLE Donation_History (donation_id I...	0 row(s) affected	0.094 sec
---	-----	----------	---	-------------------	-----------

Query 13:

```
START TRANSACTION;
INSERT INTO BOK (book_id,title,isbn,published_year)
VALUES (4,'Animal Farm','9978654321980',1945);
INSERT INTO Donation_History(donation_id,book_id,donar_name,donation_date)
VALUES(1,4,'Raj Kumar',CURDATE());
COMMIT;
```

Output: -

✓	109	14:02:19	COMMIT	0 row(s) affected
✓	110	14:02:25	START TRANSACTION	0 row(s) affected
✓	111	14:02:29	INSERT INTO BOK (book_id,title,isbn,published_...	1 row(s) affected

✓ 113 14:02:50 INSERT INTO Donation_History(donation_id,boo... 1 row(s) affected

Query 14: Index on ISBN

```
CREATE INDEX idx_books_isbn ON Bok(isbn);
```

Output :-

✓ 116 14:05:25 CREATE INDEX idx_books_isbn ON Bok(isbn) 0 row(s) affected Records: 0 Duplicates: 0 Wami... 0.125 sec

Query 15: Fast Search

```
SELECT * FROM BOK WHERE isbn = '9780743273565';
```

Output : -

	book_id	title	isbn	published_year
▶	1	The Great Bay	9780743273565	2000
*	NULL	NULL	NULL	NULL

✓ 117 14:07:50 SELECT * FROM BOK WHERE isbn = '97807432... 1 row(s) returned